Financial Data Mining and Analysis

Course: FICO3002

Semester: Spring 2026

Instructor: Dr. Liqun (Lee) Zhuge

Semester Schedule

17 weeks

“2-1” schedule: Every 2 theoretical classes + 1 coding class

One Midterm

One Group Coding Project

Semester Schedule

Week	
1	Logistic + Lecturing
2	Lecturing 1
3	Coding class 1
4	Lecturing 2
5	Lecturing 3
6	Coding class 2 (project requirements release)
7	Lecturing 4
8	Lecturing 5
9	Midterm exam
10	Lecturing 6
11	Lecturing 7
12	Coding class 3
13	Lecturing 8
14	Lecturing 9
15	Coding class 4
16	Presentation
17	Presentation

Grading

Participation 10%

Midterm exam 20%

Coding project 20%

Final exam 50%

Agenda

A quick introduction

- Why Python?
- Comparisons between R, SAS, and Python
- Textbook, errata, and code on GitHub
- Chapter 1: Python basics

Open-Source-Finance

- Three components
  - Open-software
    - Such as R, Python, Octave, and Julia
  - Open data
    - Yahoo! Finance, SEC filings, FRED
  - Open code
    - GitHub

Kane (2019)

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=966354

Which language?

SAS versus R debate

http://insidebigdata.com/2014/03/01/sas-versus-r/

SAS vs. R (vs. Python) - which tool should I learn?

https://www.analyticsvidhya.com/blog/2014/03/sas-vs-vs-python-tool-learn/

SAS vs. R (Part I)

https://thomaswdinsmore.com/2014/12/01/sas-versus-r-part-1/

SAS vs. R (Part II)

https://thomaswdinsmore.com/2014/12/15/sas-versus-r-part-two/

R or SAS: which one is the best statistical software used in medical field?

https://www.researchgate.net/post/R_or_SAS_which_one_is_the_best_statistical_software_used_in_medical_field

R vs SAS (Comparison and Opinion)

http://www.learnanalytics.in/blog/?p=9

R vs SAS, why is SAS preferred by private companies?

http://stats.stackexchange.com/questions/33780/r-vs-sas-why-is-sas-preferred-by-private-companies

Objectives of this course

- Appy Python to finance such as Ratio analysis (Financial Statement Analysis), Portfolio Theory, CAPM, Fama-French 3-, 4-, 5-factor Models, Monte Carlo simulation, Options Theory, VaR, spread estimation from daily data, from high-frequency data (optional)

- Learn and apply Python to finance

- Focus on publicly available econ/fin data

- Optional : Use CRSP/Compustat/TAQ databases

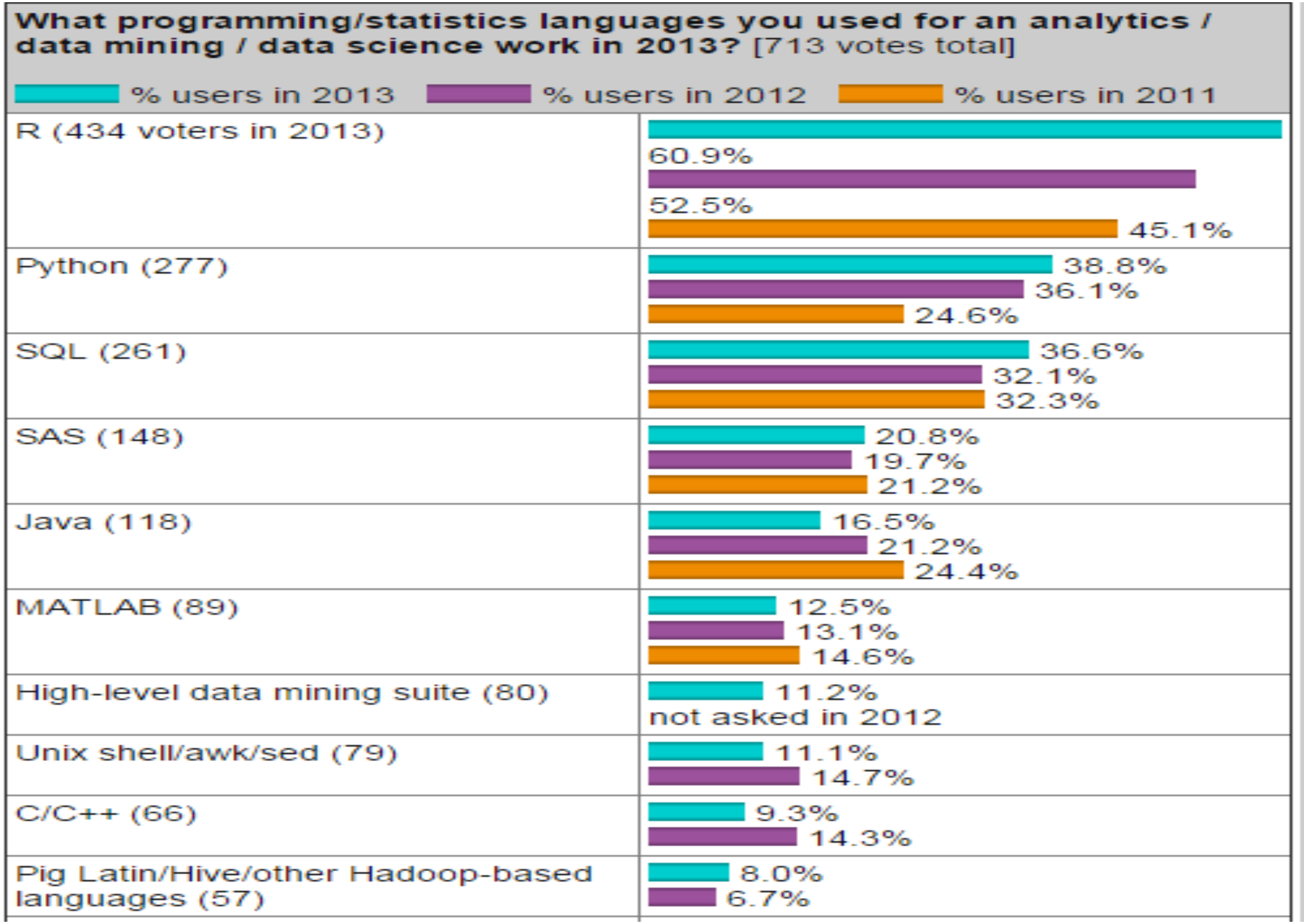
  Note): Can use CRSP or TAQ to finish their term projects.

Why Python?

Comparison of data analysis packages: R, Matlab, SciPy, Excel, SAS, SPSS, Stata

Posted on February 23, 2009

Name	Advantages	Disadvantages	Open source?	Typical users
R	Library support; visualization	Steep learning curve	Yes	Finance; Statistics
Matlab	Elegant matrix support; visualization	Expensive; incomplete statistics support	No	Engineering
SciPy/NumPy/Matplotlib	Python (general-purpose programming language)	Immature	Yes	Engineering
Excel	Easy; visual; flexible	Large datasets	No	Business
SAS	Large datasets	Expensive; outdated programming language	No	Business; Government
Stata	Easy statistical analysis		No	Science
SPSS	Like Stata but more expensive and worse			


What programming/statistics languages you used for an analytics / data mining / data science work in 2013? [713 votes total]
% users in 2013
% users in 2012
% users in 2011
R (434 voters in 2013)
60.9%
52.5%
45.1%
Python (277)
38.8%
36.1%
24.6%
SQL (261)
36.6%
32.1%
32.3%
SAS (148)
20.8%
19.7%
21.2%
Java (118)
16.5%
21.2%
24.4%
MATLAB (89)
12.5%
13.1%
14.6%
High-level data mining suite (80)
11.2%
not asked in 2012
Unix shell/awk/sed (79)
11.1%
14.7%
C/C++ (66)
9.3%
14.3%
Pig Latin/Hive/other Hadoop-based languages (57)
8.0%
6.7%

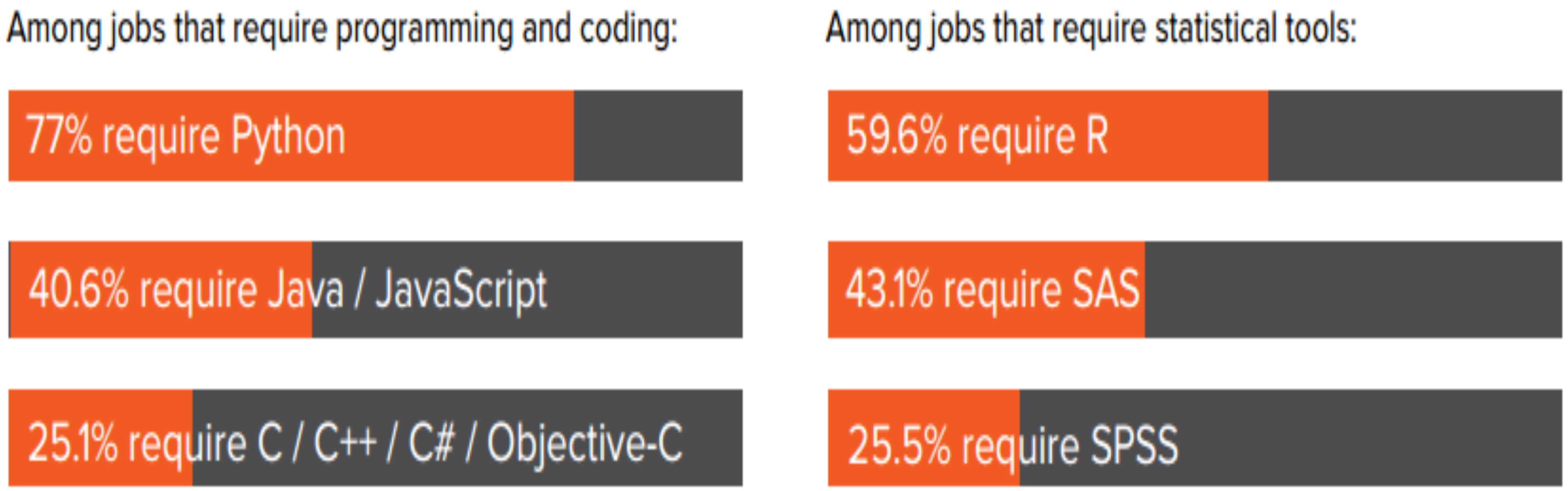
A quick comparison (5 is the best)

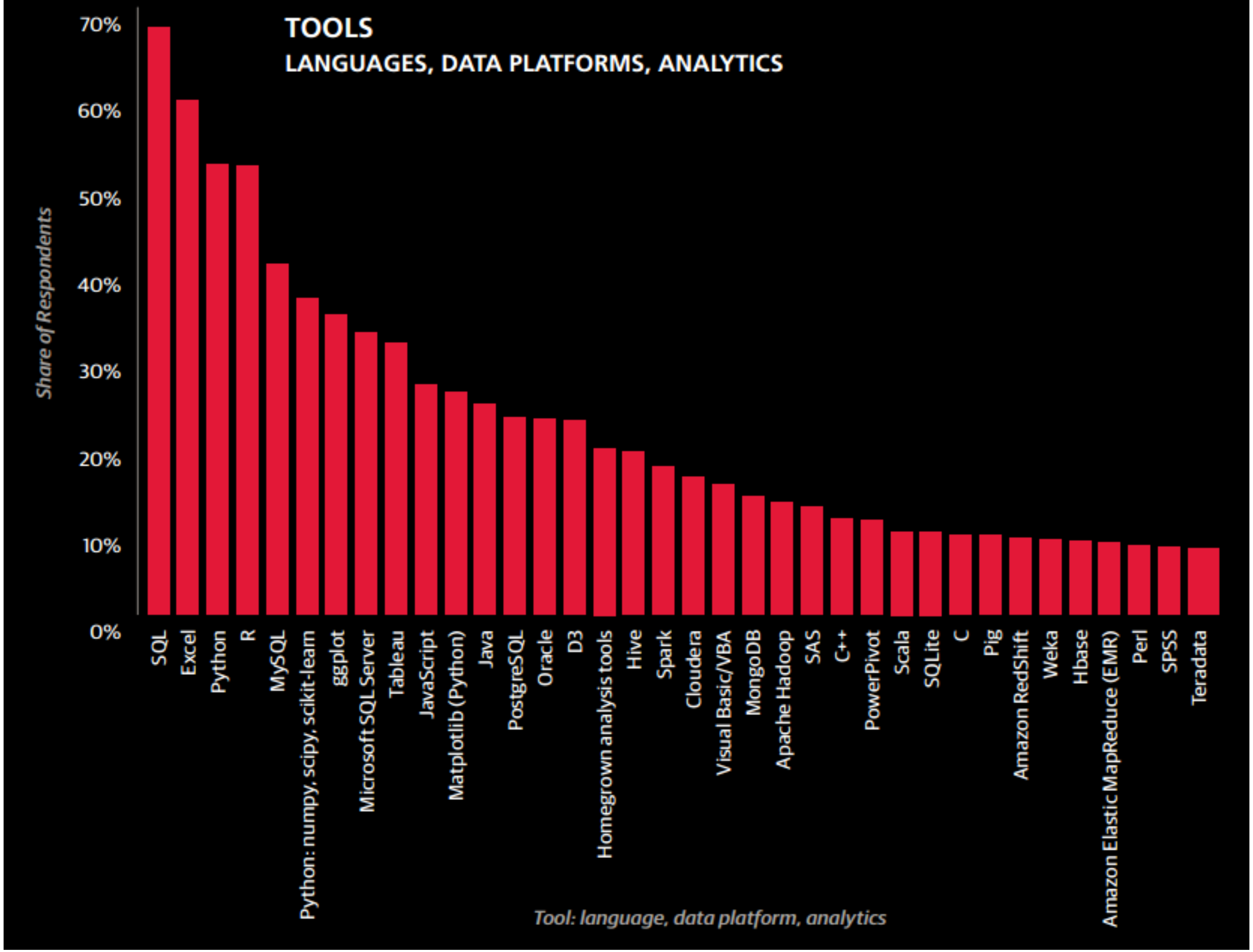
Parameter	SAS	R	Python
Availability / Cost	2	5	5
Ease of learning	4.5	2.5	3.5
Data handling capabilities	4	4	4
Graphical capabilities	3	4.5	4
Advancements in tool	4	4.5	4
Job scenario	4.5	3.5	2.5
Customer service support and Community	4	3.5	3

http://www.analyticsvidhya.com/blog/2014/03/sas-vs-vs-python-tool-learn/

	R	Python	SAS	Matlab
Cost	5	5	2	2
Easy of learning	5	4	2	2
Data handling	4	4	5	2
Graph	4.5	4	3	4.5
Current job	4	4	5	5
Future trend	5	5	3	4
Support	2	2	5	5

http://visit.crowdflower.com/rs/416-ZBE-142/images/Crowdflower_Data_Scientist_Survey2015.pdf





https://duu86o6n09pv.cloudfront.net/reports/2015-data-science-salary-survey.pdf

Why Python?

- Python has become a dominant language in finance, offering a versatile and powerful toolset for various financial applications. Renowned for its <u>readability and extensive library support</u>, Python facilitates the development of sophisticated financial models, quantitative analysis, and algorithmic trading strategies.

Why Python (2)

- Libraries such as NumPy and Pandas enable efficient data manipulation and analysis, while Matplotlib and Seaborn provide robust visualization capabilities.
- Python's integration with Jupyter notebooks allows for interactive and collaborative analysis, making it an ideal choice for financial analysts, quants, and researchers.

Why Python (3)

- Furthermore, the adoption of Python in finance is underscored by its compatibility with machine learning frameworks like TensorFlow and scikit-learn, enabling the implementation of predictive models and risk management solutions.
- With its accessibility, community support, and a rich ecosystem of libraries, Python has emerged as a go-to language for professionals in the finance industry seeking to streamline quantitative analysis, financial modeling, and data-driven decision-making.

Anaconda installation

Method I: install Anaconda, which includes Python:

a) http://anaconda.org

https://www.anaconda.com/products/individual

b)Choose appropriate software

Windows, Python 3.8

64-Bit Graphical Installer (457 MB)

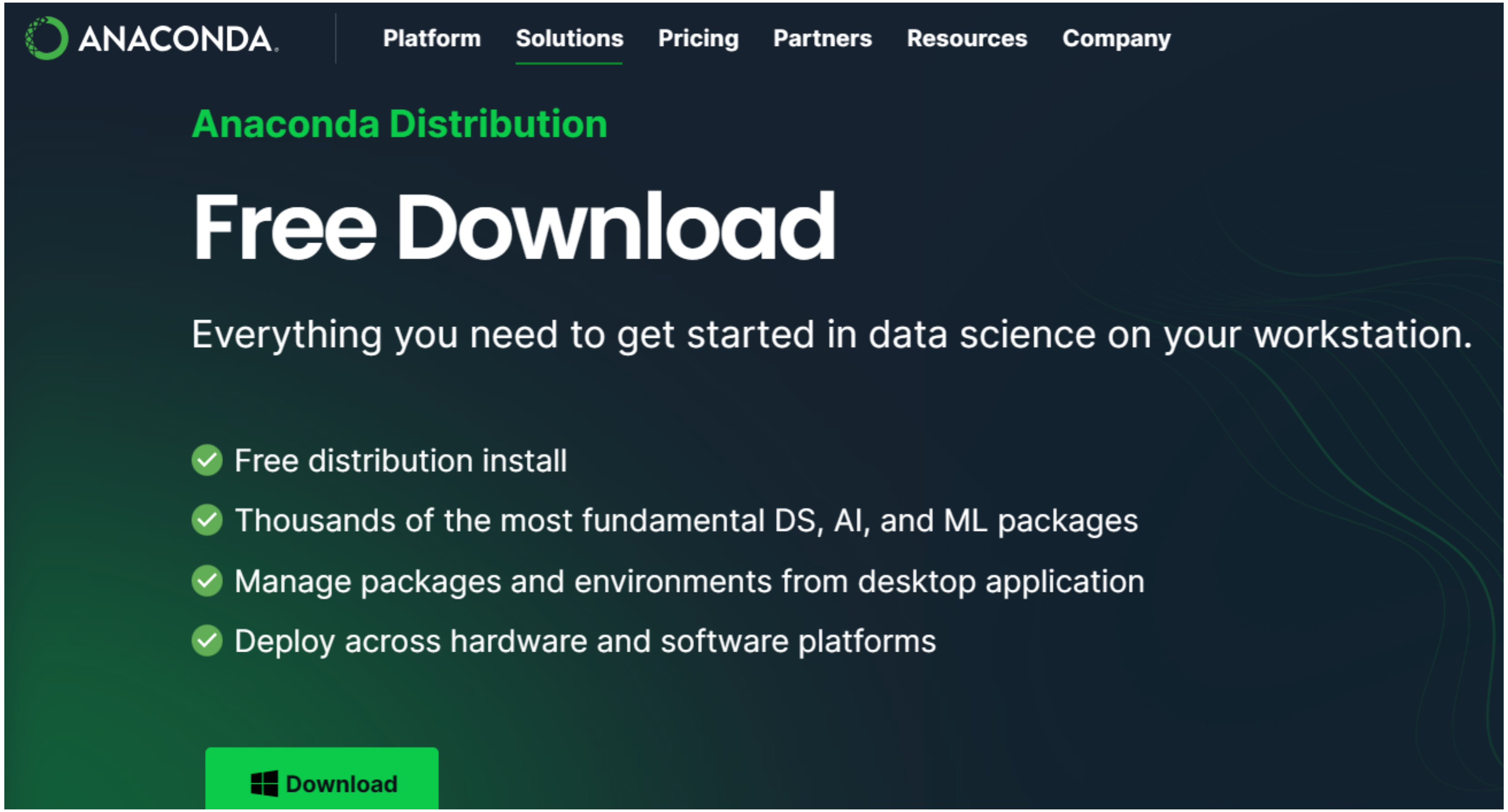
32-Bit Graphical Installer (403 MB)

MacOs, Python 3.8

64-Bit Graphical Installer (435 MB)

64-Bit Command Line Installer (428 MB)

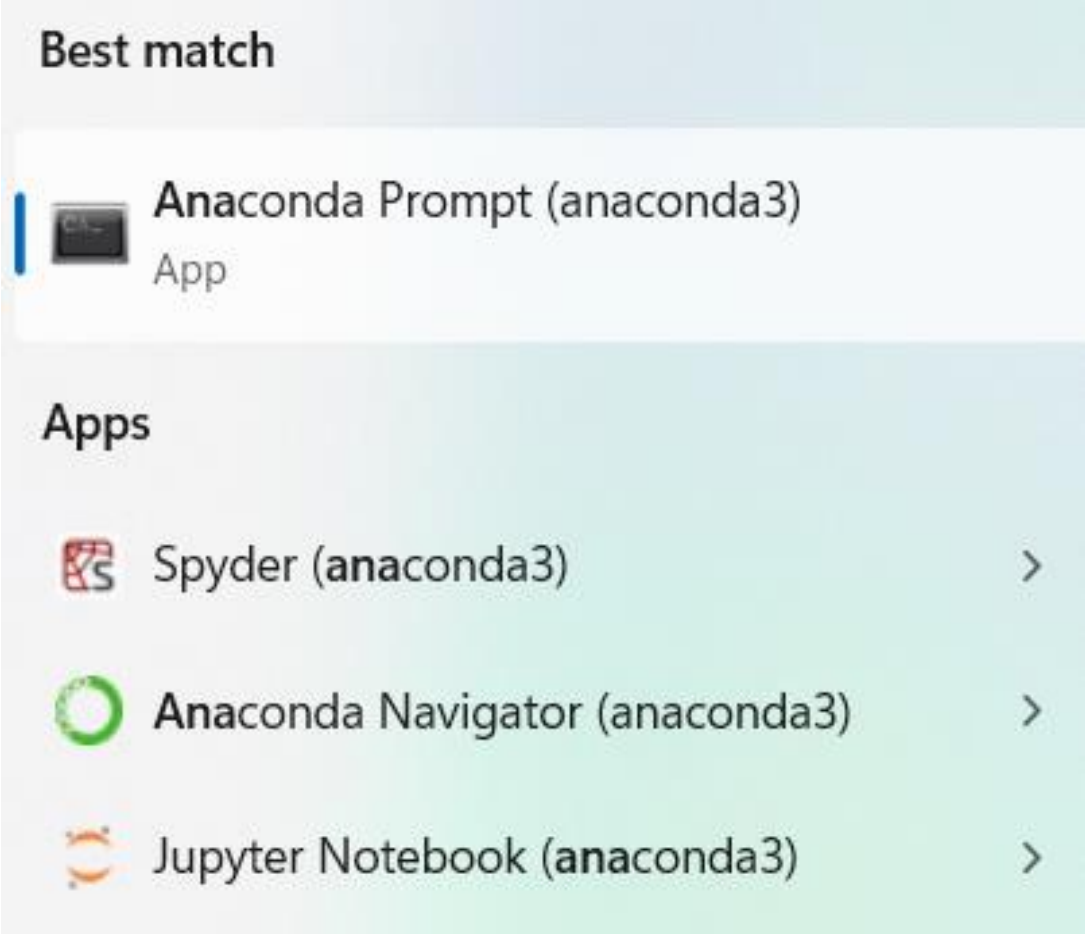
Anaconda installation



How to launch Python?

- Method I: Spyder
- Method II: Jupyter Notebook
- Method III: Jupyter Lab
- Method IV: Colab of Google
- Method V: launch Python

- Searching for Anaconda,
  See the right image

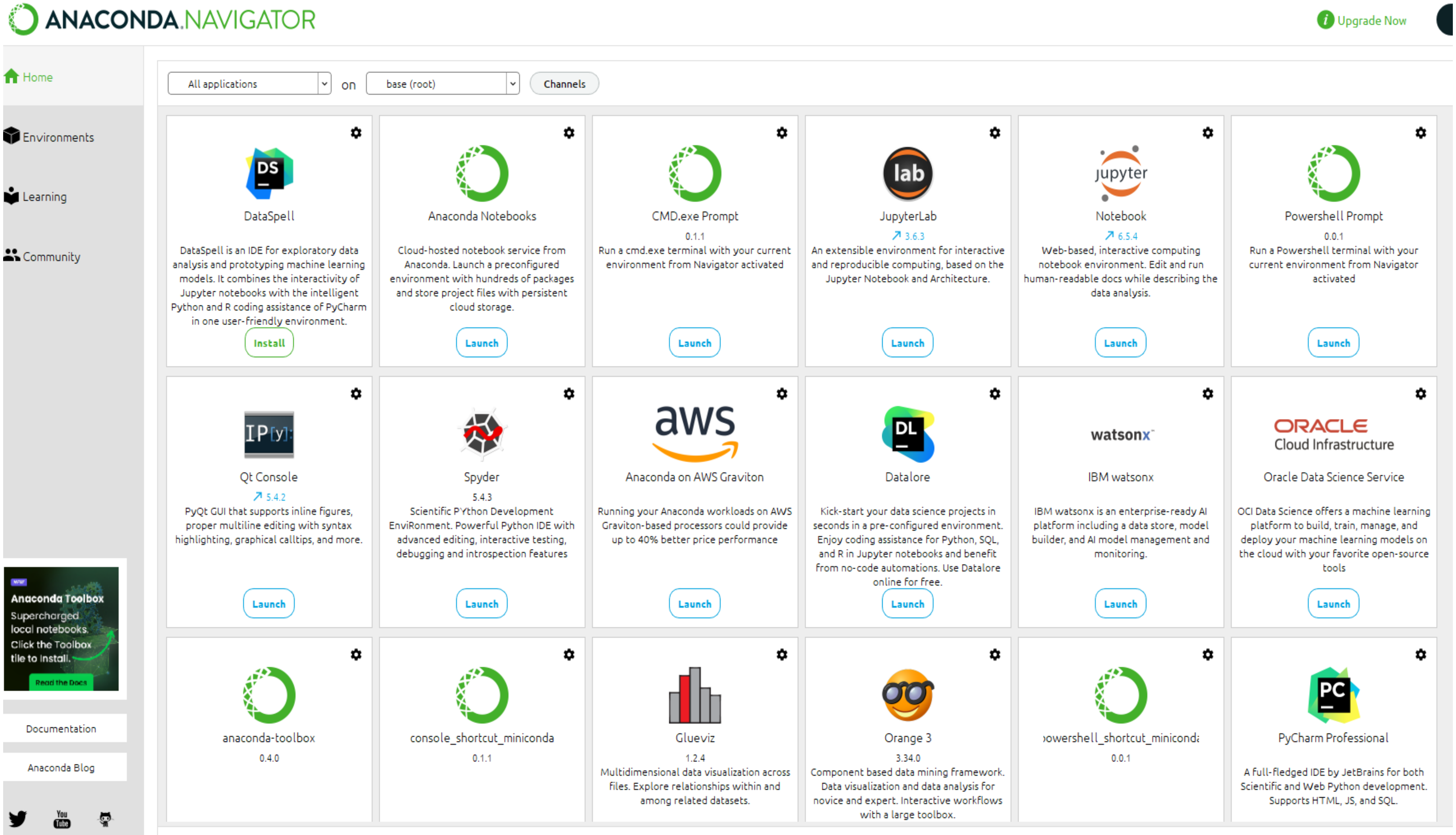


After Anaconda was installed

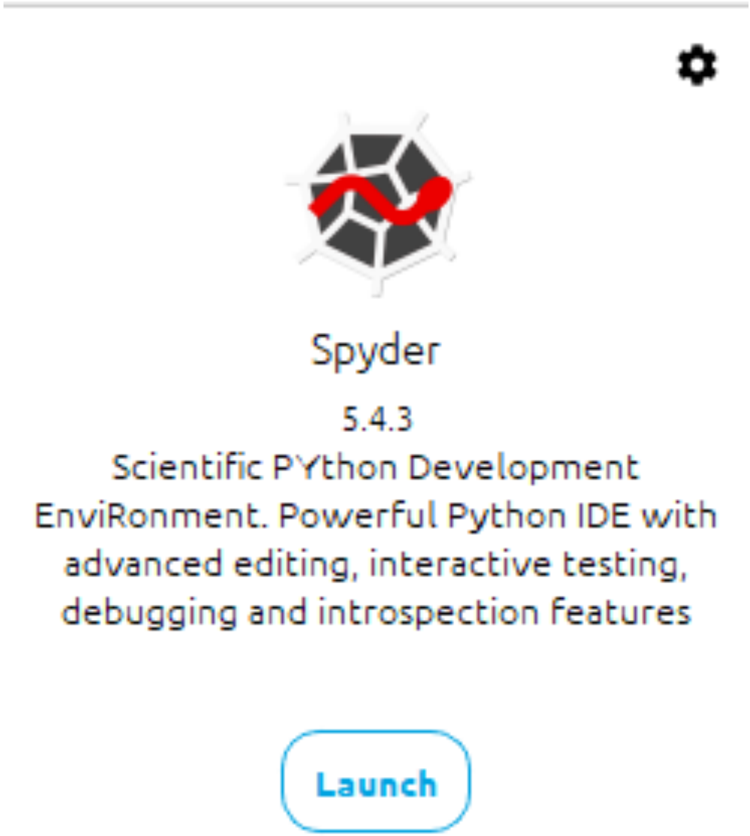
- We can launch Spyder, Jupyter Notebook, and Jupyter Lab via Anaconda Navigator

- We launch Anaconda Navigator

  See the next slide.

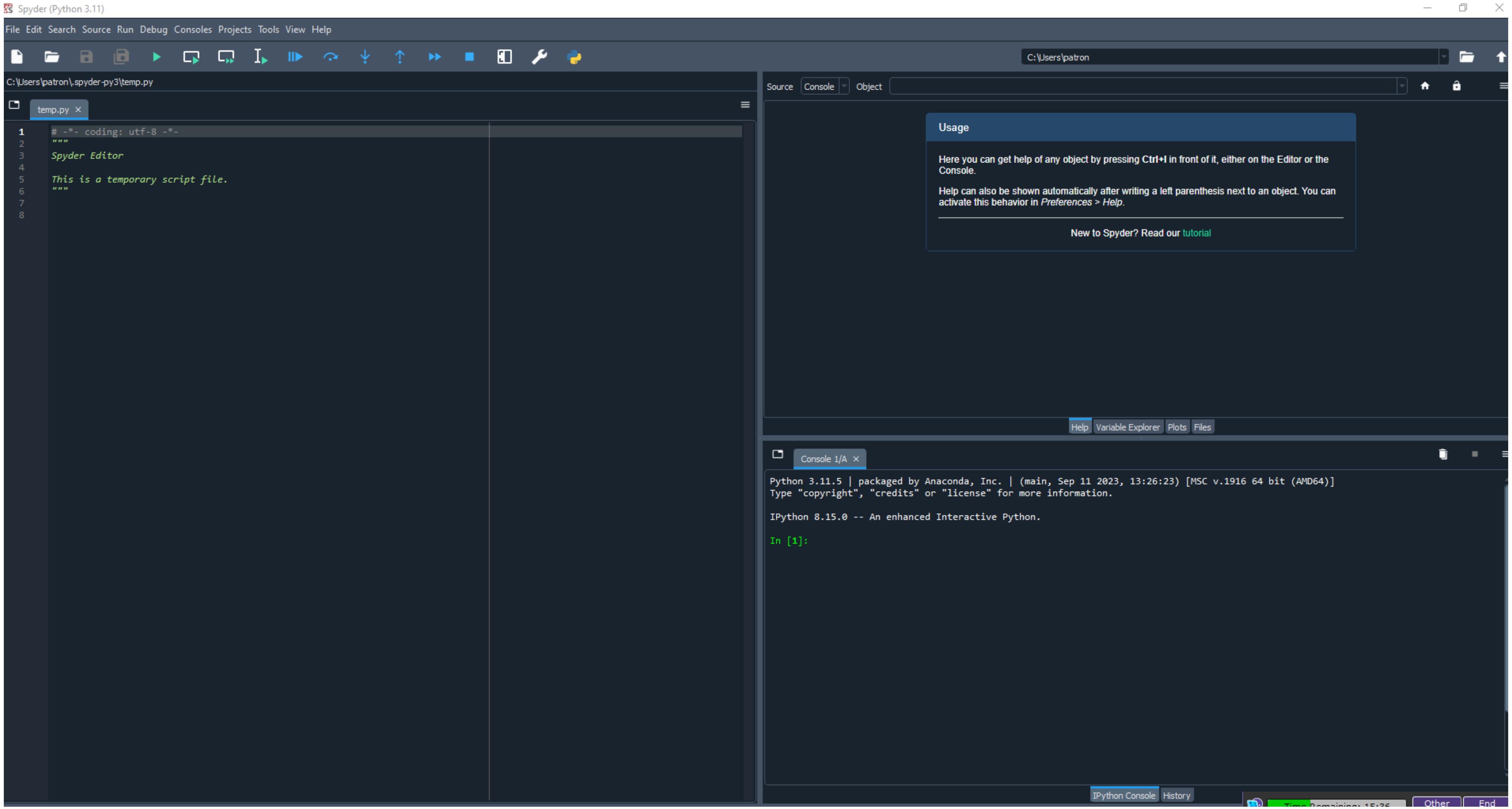

ANACONDA.NAVIGATOR
Upgrade Now
Home
Environments
Learning
Community
All applications
on
base (root)
Channels
DataSpell
DataSpell is an IDE for exploratory data analysis and prototyping machine learning models. It combines the interactivity of Jupyter notebooks with the intelligent Python and R coding assistance of PyCharm in one user-friendly environment.
Install
Anaconda Notebooks
Cloud-hosted notebook service from Anaconda. Launch a preconfigured environment with hundreds of packages and store project files with persistent cloud storage.
Launch
CMD.exe Prompt
0.1.1
Run a cmd.exe terminal with your current environment from Navigator activated
Launch
JupyterLab
3.6.3
An extensible environment for interactive and reproducible computing, based on the Jupyter Notebook and Architecture.
Launch
Notebook
6.5.4
Web-based, interactive computing notebook environment. Edit and run human-readable docs while describing the data analysis.
Launch
Powershell Prompt
0.0.1
Run a Powershell terminal with your current environment from Navigator activated
Launch
Qt Console
5.4.2
PyQt GUI that supports inline figures, proper multiline editing with syntax highlighting, graphical calltips, and more.
Launch
Spyder
5.4.3
Scientific PYthon Development EnviRonment. Powerful Python IDE with advanced editing, interactive testing, debugging and introspection features
Launch
aws
Anaconda on AWS Graviton
Running your Anaconda workloads on AWS Graviton-based processors could provide up to 40% better price performance
Launch
Datalore
Kick-start your data science projects in seconds in a pre-configured environment. Enjoy coding assistance for Python, SQL, and R in Jupyter notebooks and benefit from no-code automations. Use Datalore online for free.
Launch
watsonx
IBM watsonx
IBM watsonx is an enterprise-ready AI platform including a data store, model builder, and AI model management and monitoring.
Launch
ORACLE Cloud Infrastructure
Oracle Data Science Service
OCI Data Science offers a machine learning platform to build, train, manage, and deploy your machine learning models on the cloud with your favorite open-source tools
Launch
Anaconda Toolbox
Supercharged local notebooks.
Click the Toolbox tile to install.
Read the Docs
Documentation
Anaconda Blog
anaconda-toolbox
0.4.0
console_shortcut_miniconda
0.1.1
Glueviz
1.2.4
Multidimensional data visualization across files. Explore relationships within and among related datasets.
Orange 3
3.34.0
Component based data mining framework. Data visualization and data analysis for novice and expert. Interactive workflows with a large toolbox.
›owershell_shortcut_miniconda
0.0.1
PyCharm Professional
A full-fledged IDE by JetBrains for both Scientific and Web Python development. Supports HTML, JS, and SQL.

How to launch Spyder?



It is a good idea to copy the path to your desktop.

Spyder’s panels



Use the Python console

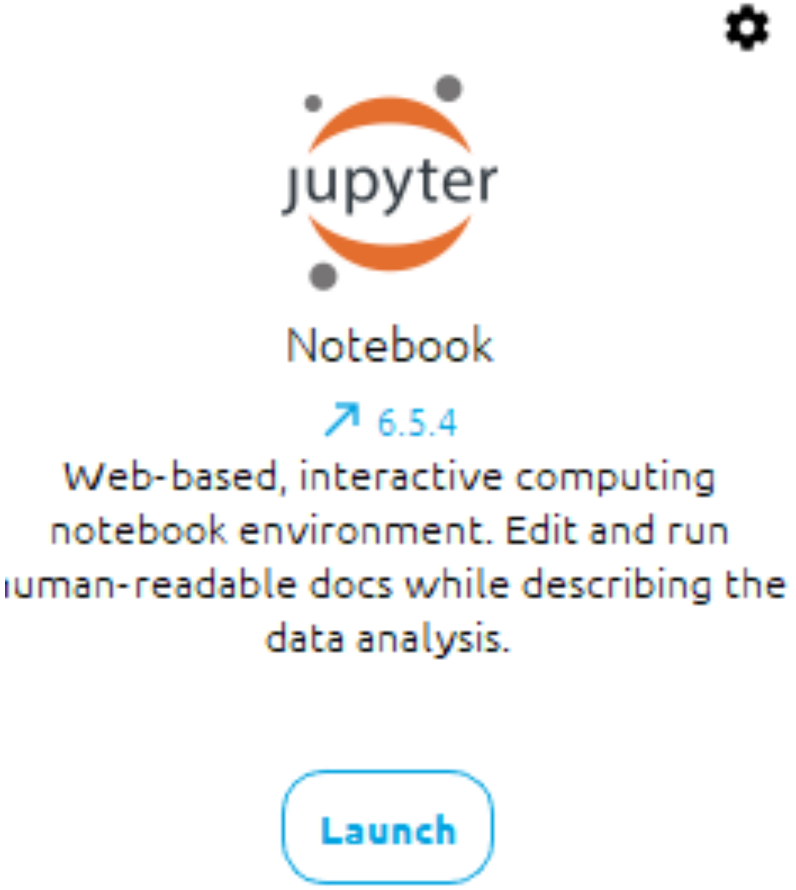
```
Console 1/A

Python 3.11.5 | packaged by Anaconda, Inc. | (main, Sep 11 2023, 13:26:23) [MSC v.1916 64
bit (AMD64)]
Type "copyright", "credits" or "license" for more information.

IPython 8.15.0 -- An enhanced Interactive Python.

In [1]: pv=100

In [2]: r=0.1

In [3]: n=4

In [4]: pv*(1+r)**n
Out[4]: 146.41000000000005

In [5]:
```

Could not launch Spyder via Anaconda

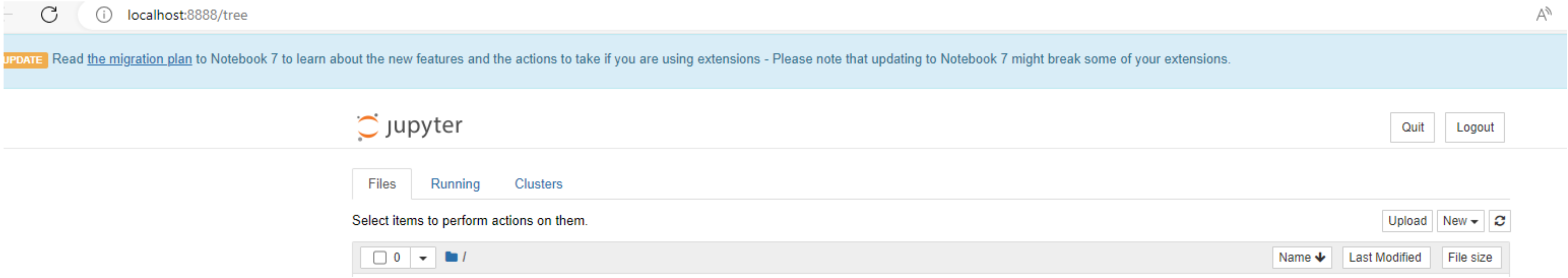
- Download Spyder separately at
- https://www.spyder-ide.org/

How do we launch Jupyter Notebook?

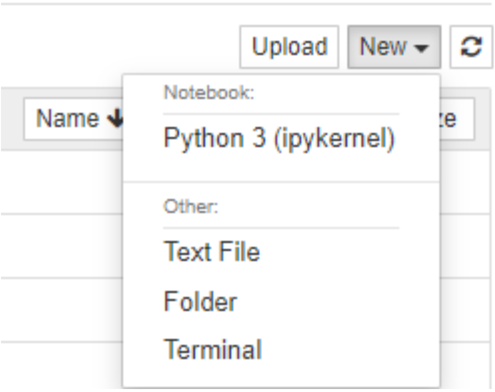
From Anaconda Navigator



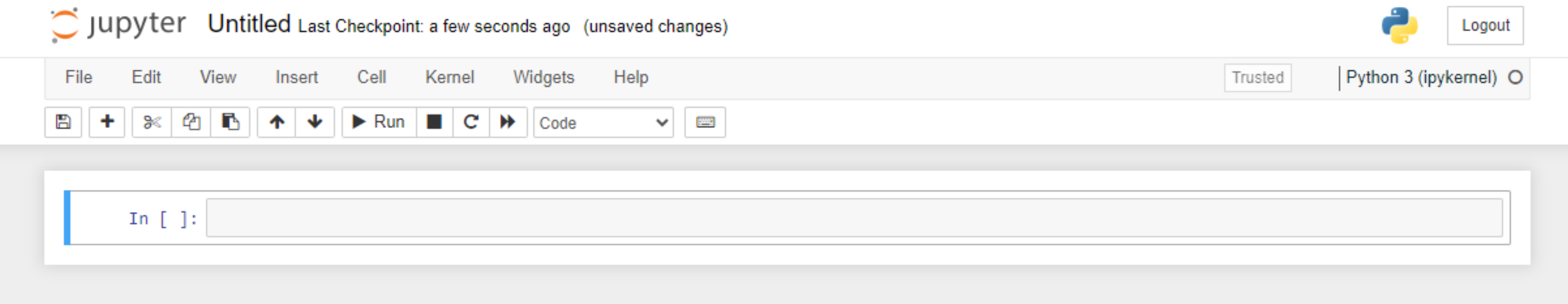
How to we launch Jupyter Notebook



Click “New”, then choose “Python 3”, shown below.

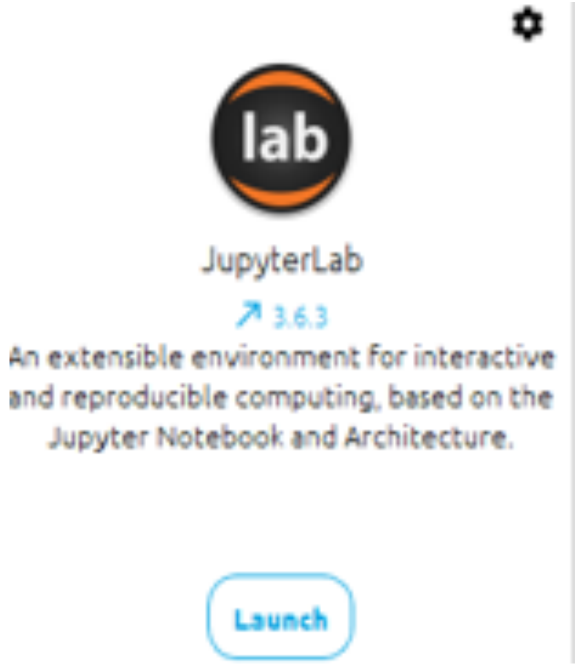


Now we can enter our code (shown below)



How to we launch Jupyter Lab

From Anaconda
Navigator

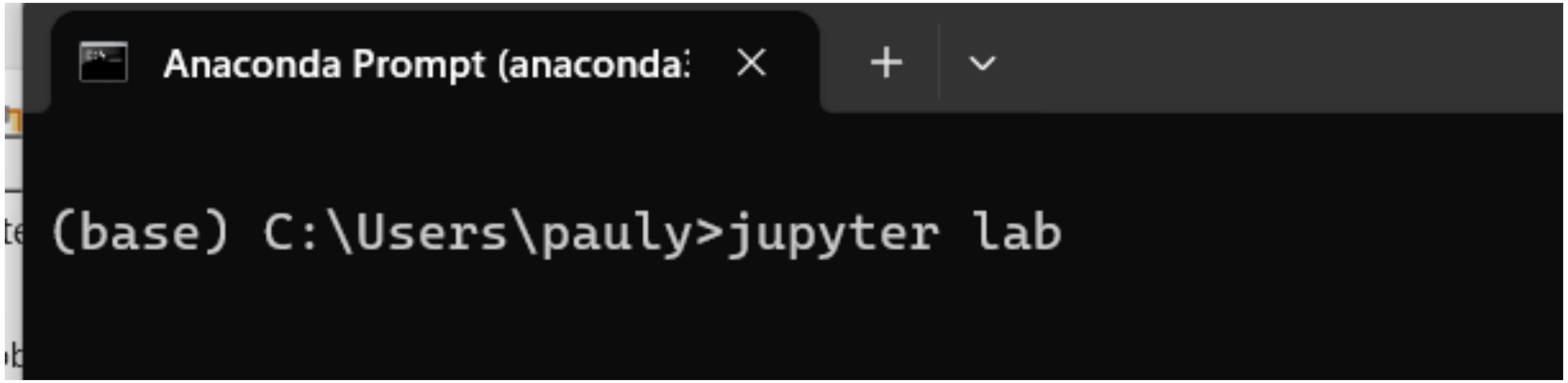


Method 2 to launch Jupyter Lab

Step 1: launch Anaconda Prompt

Step 2: type

jupyter lab



If your browser could not launch it directly, see below. Then copy the link

Untitled | week1 | week2 | Howeve | c12_15_ | Untitled | Campus_vist3 | locatior | Certainl | chapter_1 | allChapters | jpserv

File Edit View

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta http-equiv="refresh" content="1;url=http://localhost:8888/lab?token=1706efcf8d2cd9c28a4c027a76e5062044be1d7c96174020" />
    <title>Opening Jupyter Notebook</title>
</head>
<body>

<p>
    This page should redirect you to Jupyter Notebook. If it doesn't,
    <a href="http://localhost:8888/lab?token=1706efcf8d2cd9c28a4c027a76e5062044be1d7c96174020">click here to go to Jupyter</a>.
</p>

</body>
</html>
```

Spyder vs. Jupyter Notebook

Choosing the Right Tool for the Job

- **Spyder is a "Workbench"**
  - Built for **building**.
  - Like a carpenter's bench: messy, full of tools, focused on the raw construction of the product.
  - *Best for:* Writing logic, fixing bugs, and creating reusable code.

Spyder vs. Jupyter Notebook

Choosing the Right Tool for the Job

- **Jupyter is a "Lab Journal"**
  - Built for **storytelling**.
  - Like a scientist's notebook: mixes data, observations, charts, and conclusions in one place.
  - *Best for:* Explaining your thought process and showing results instantly.

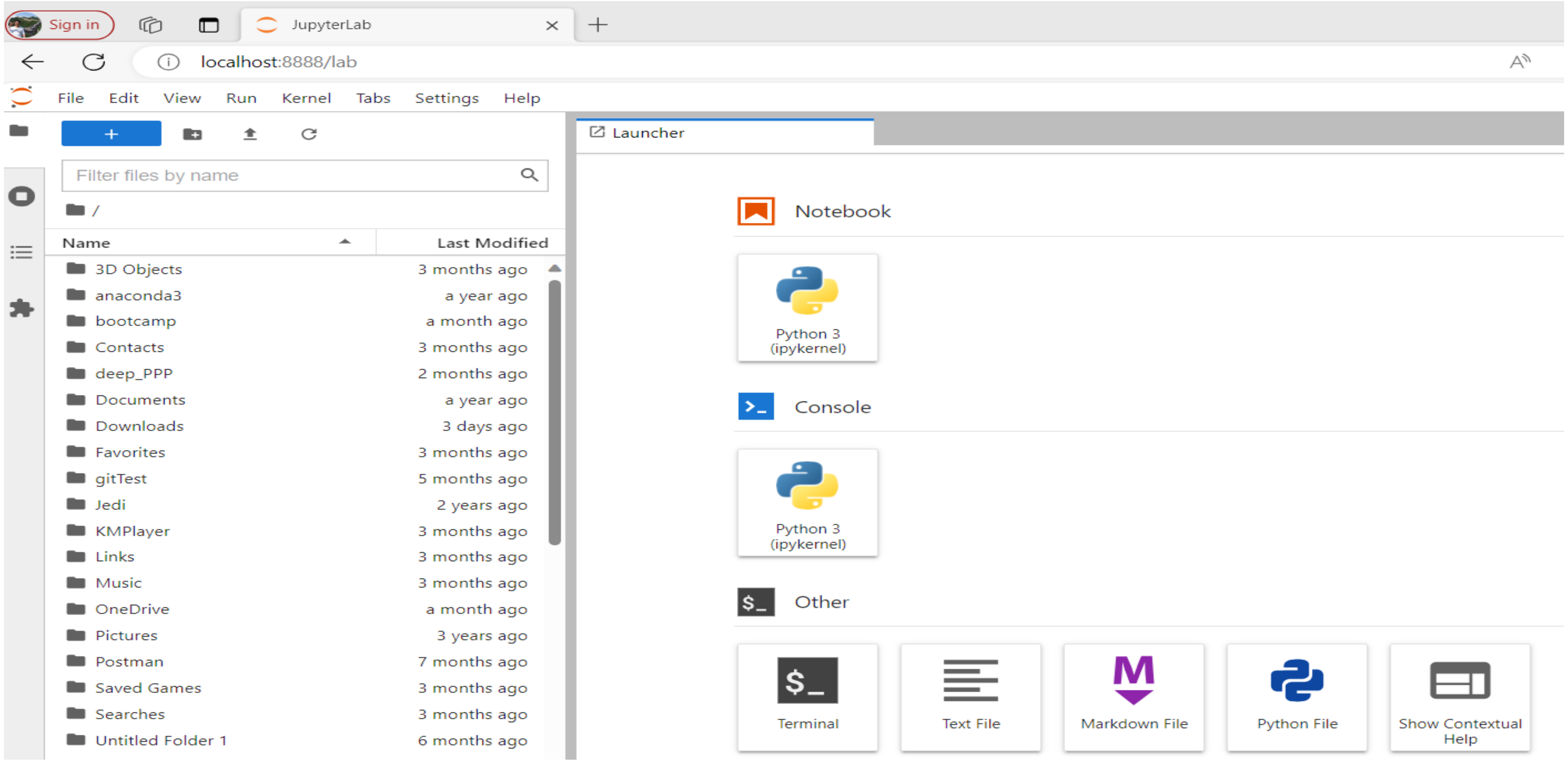
When to use Spyder?

- **Use Spyder when you are "Engineering":**
- **Heavy Coding:** Writing long scripts (over 100 lines) or complex functions.
- **Debugging:** You need to pause code line-by-line to find an error.
- **Variable Inspection:** You need to see the current state of all your data variables at once.
- **Consistency:** You want to run the exact same script multiple times without manual input.

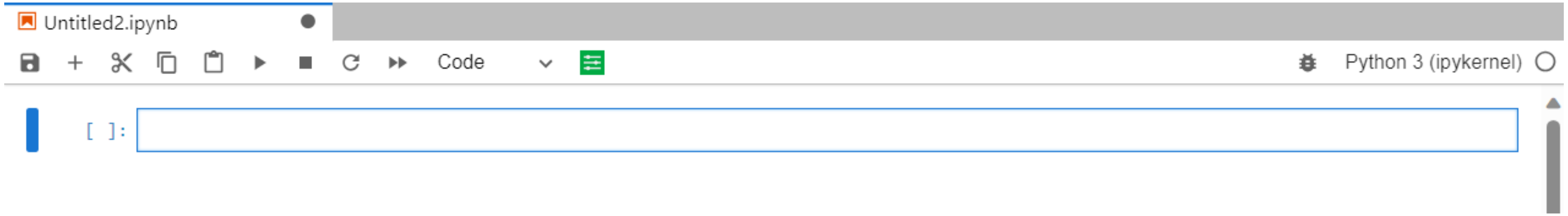
When to use Jupyter?

- **Use Jupyter when you are "Exploring":**
- **Data Analysis (EDA):** You need to tweak a graph and see the change immediately.
- **Reporting:** You need to write text (markdown) to explain what a specific result means.
- **Documentation:** You want to combine code, math equations, and charts in a single shareable file.
- **Teaching/Sharing:** You want to walk someone else through your logic step-by-step.

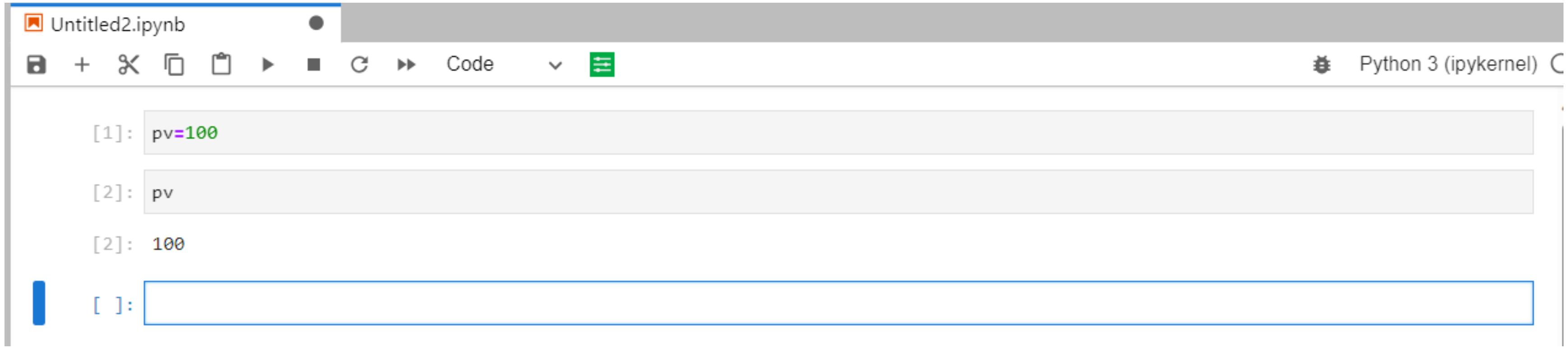
We will have the following interface



After choosing Python 3



We can use the combination of shift-enter to execute a code



Python 2 is unsupported anymore (Since Jan 2020)

Chapter 1: Python basics

- Anaconda installation
- Launching Python
  - Spyder
  - Jupyter Notebook
  - Jupyter Lab
- Python basics

Python basics

1) >>>   # '>>>' is the Python prompt

2) Python is case-sensitive

```
>>>x=10
>>>X
   name 'X' is not defined
```

3) Alignment is critical for Python

4) Assign a value

```
>>>y=[1,2,5]
```

5) count from 0

```
>>>y=[1,2,5]
 y[0]
 1
```

[1,2,5] vs. (1,2,5)

```
>>>x=[1,2,5]
 >>>type(x)
   Out[38]: list
 >>>x[0]=10       # you can change its value

 >>>y=(1,2,5)
 >>>type(y)
   Out[39]: tuple
 >>>y[0]=10
   Traceback (most recent call last):
     File "<ipython-input-43-f39ee00d3d8d>", line 1, in <module>
     y[0]=10
   TypeError: 'tuple' object does not support item assignment
```

The primary difference between a list and a tuple in Python is **mutability**.

- **Lists are Mutable:** You can change, add, or remove items after the list is created.
- **Tuples are Immutable:** Once created, they cannot be changed. You cannot add, remove, or modify elements.

e, pi and log10

```
import math
math.e

math.pi

math.log10
```

String variables and .upper() and .lower()

```
x='Mary'

x.upper()
```

Two types of comments

- Type 1: anything after #

- Type 2: Two pair of triple double quotation marks

"""

This part is a multiple-line comment

"""

Assume that we want to calculate the square root of 3.

```
>>>sqrt(3)
Traceback (most recent call last):
  File "<ipython-input-1-2f71726bed4c>", line 1, in <module>
    sqrt(3)
NameError: name 'sqrt' is not defined

import math
math.sqrt(3)
Out[2]: 1.7320508075688772

from math import sqrt  # Alternatively, we can use
sqrt(3)
Out[3]: 1.7320508075688772
```

import math module

```
import math
x=math.sqrt(3)
print(x)
```

Print all columns

In Python, pd.set_option() is a function from the **Pandas** library

It is used to **configure and customize the display behavior** of Pandas data structures, such as DataFrames and Series. It helps you control how your data looks when printed to the console or displayed in a Jupyter Notebook.

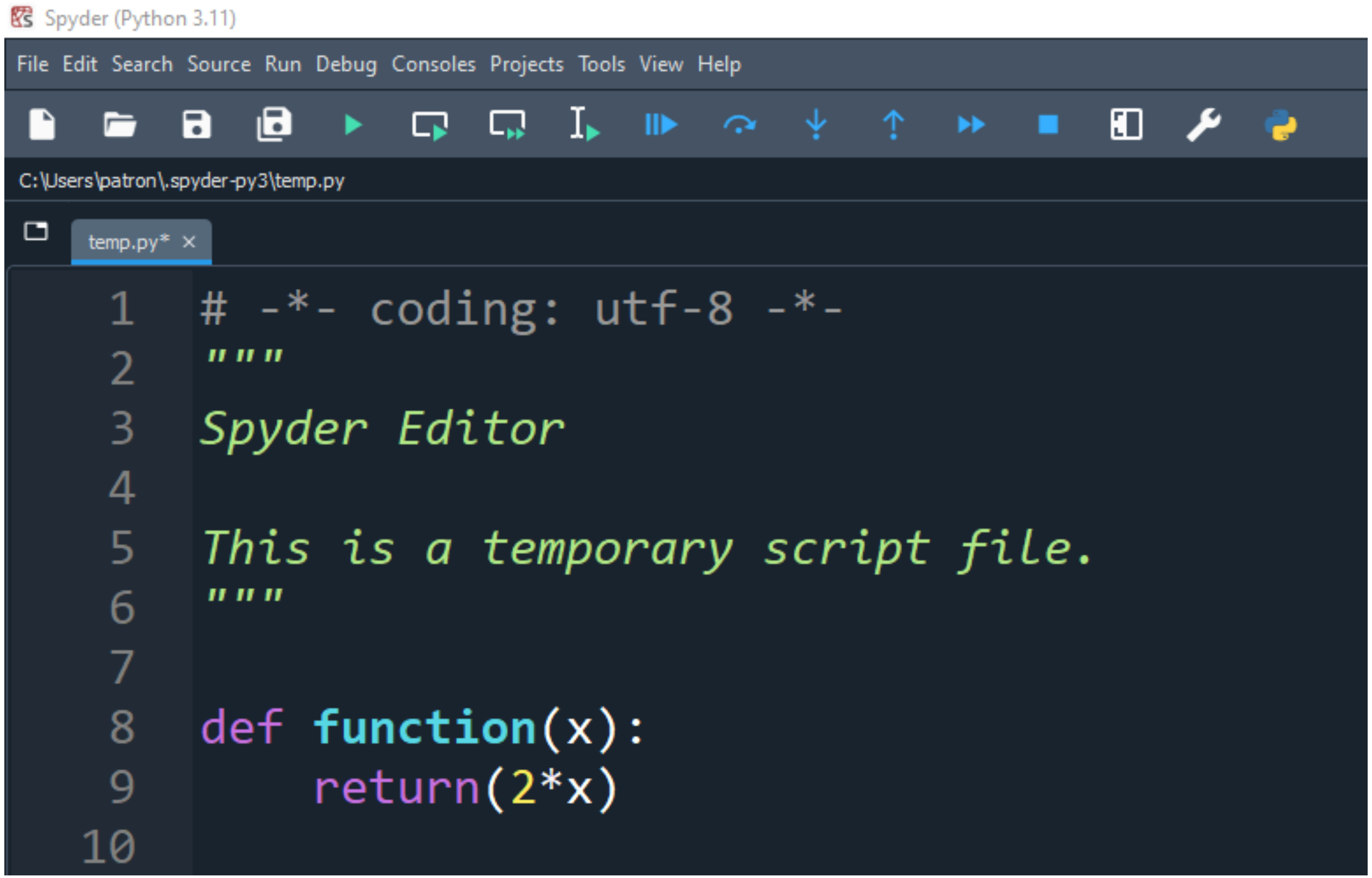
```
pd.set_option('display.max_rows', 500)
pd.set_option('display.max_columns', 500)
pd.set_option('display.width', 1000)

# return to the original setting
#----------------------------------
pd.reset_option('max_columns')
pd.reset_option('display.max_rows')
pd.reset_option('display.width')
```

The simplest Python functions

```
def dd(x):
   return(x*2)
```

```
1) def is the keyword
2) dd is the function name
3) x is the input
4) alignment is important
5) return() returns the result
6) x*2 will double an input value
```

Using Spyder's editor



Future value function

$fv =$pv(1+r)^n

fv is the future value
Pv is the present value
r is the rate
N is the number of periods

```
def fv_f(pv,r,n):
    return pv*(1+r)**n

fv=fv_f(100,0.1,2)
print(fv)
```

Three ways to input values

- Method I: the meanings of input variables depend on their order
- Method II: the meanings of input variables depend on their keywords
- Method III: mixed ones

In-class exercise #3: write a pv_f() function

$pv =$`fv/(1+r)^n`

Method I: input values depend on order i.e., their positions

```
In [8]: fv=fv_f(100,0.1,1)
   ...: print(fv)
110.00000000000001
```

```
In [9]:
   ...: fv=fv_f(0.1,100,1)
   ...: print(fv)
10.100000000000001
```

Method II: input values depend on their keywords

```
In [10]: fv1=fv_f(100,0.1,1)
    ...: fv2=fv_f(r=0.1,pv=100,n=1)
    ...: fv3=fv_f(n=1,r=0.1,pv=100)
    ...: print(fv1)
    ...: print(fv2)
    ...: print(fv3)
110.00000000000001
110.00000000000001
110.00000000000001

In [11]:
```

Keyboard short-cut

- run current line --> F9

- run selected several lines --> F9

- Run cell   --> Ctrl+Return

- Note that if you prefer to use Ctrl+Return,
-     you can go to Tools -> Preferences ->
-     Keyboard shortcuts. Search for 'run selection',
-     double click and set Ctrl+Return as the 'New shortcut'

In-class exercise #4: add comments (helpt) to your pv_f() function

$pv =$`fv/(1+r)^n`

In an R-assisted learning environment

- First, download and install R

To install R, we have the following steps:

Step 1: go to http://r-project.org

Step 2: click "CRAN" on the left

Step 3: choose a server nearby

Step 4: choose appropriate type such as Windows, Mac

Step 5: download "Base"

15 weeks' extra teaching materials

- source('http://datayyy.com/p4f.txt')

- Note that the above file contains the first line below. Each week instructors can modify it for different week's material. Thus, students are given just one line of R code.
- source('http://datayyy.com/p4f/week1.txt')
- source('http://datayyy.com/p4f/week2.txt')
- source('http://datayyy.com/p4f/week3.txt')

- source('http://datayyy.com/p4f/week14.txt')
- source('http://datayyy.com/p4f/week15.txt')